

Use-Case Flow Specification

Problem Statement #56 :- Peer Skill Exchange & Mentorship Network (PES1UG24CS244)

Use Case ID / Name	UC-04: Book Skill Exchange Session
Actors	Learner Member (primary), Skill Mentor
Related Requirements	FR-001, FR-003 (include: Check Time-Credit Balance), NFR-001 (include: Sync Session to Calendar)

Preconditions

- The Learner Member has an active, verified account and is logged in.
- The Skill Mentor's profile and availability calendar are published and up to date.

Postconditions (Success)

- The session slot is reserved and blocked on both the Mentor's and the Learner's calendars.
- Both parties receive a booking confirmation notification.
- (After the session takes place) 1 hour of time-credit is transferred from the Learner to the Mentor once both sign off, and both members are prompted to rate each other.

Main Success Scenario

1. Learner Member logs in and searches for a skill category (e.g., "Guitar Lessons") using UC-02 (Search for Skill / Browse Mentors).
2. System displays a list of matching Skill Mentors along with their availability calendars.
3. Learner Member selects a preferred Skill Mentor and a specific session time slot.
4. System includes UC-03 (Check Time-Credit Balance) to confirm the Learner holds at least 1 hour of time-credit.
5. System reserves the session slot and includes UC-05 (Sync Session to Calendar) to push the event to both members' calendars via iCal.
6. System sends a booking confirmation to both the Learner Member and the Skill Mentor.
7. The scheduled skill-exchange session takes place outside the system (in person or via video call).
8. After the session, both the Learner Member and the Skill Mentor open the app and confirm completion (sign-off) via UC-06.
9. System transfers 1 hour of time-credit from the Learner's balance to the Mentor's balance and prompts both members to rate each other (UC-07).
10. Use case ends successfully.

Alternate Flow — 4a. Insufficient Time-Credit Balance

- 4a1.** System displays an "Insufficient time credits" message and blocks the booking.
- 4a2.** Learner Member is prompted to either offer a teaching session of their own to earn credits, or select a shorter session that fits their available balance.
- 4a3.** If the Learner Member does not take either action, the booking attempt is cancelled and the use case ends without a reserved slot.